

Photosynthetic refixation in branches of Western White Pine

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Summary

1. We investigated the functional controls on light-dependent refixation of respired CO₂ in photosynthetic bark of Western White Pine (*Pinus monticola* Dougl. ex D. Don) branches.
2. Gross photosynthesis increased with increasing intensity of photosynthetically active radiation (PAR) to > 2000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ and with increasing bark surface temperature from 15 to > 45 °C. Maximum observed refixation rates were $76 \pm 3\%$ (mean $\pm$ SE) of dark respiration.
3. At a given temperature and irradiance, dark respiration rate was the physiological parameter that correlated most strongly with gross photosynthesis ($R = 0.89$, $P < 0.0001$, $n = 43$). Dark respiration rates varied 20-fold among branches. Over this range, refixation averaged $55 \pm 2\%$ of dark respiration at 25 °C and 1000 $\mu\text{mol PAR m}^{-2} \text{s}^{-1}$. However, refixation was not correlated with internal CO₂ concentrations in illuminated branches; these concentrations remained fairly constant.
4. Specific bark area, bark conductance to water vapour, bark chlorophyll *a/b* ratio, and bark nitrogen concentration varied markedly with canopy depth in mature trees.
5. Results are summarized in a simple model of refixation driven by light, temperature and dark respiration rate. The model may prove useful in assessing the quantitative significance of refixation in the carbon balance of forest trees.

Key-words: Carbon balance, corticular photosynthesis, *Pinus monticola*, woody-tissue respiration

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Introduction

Productivity in forest trees is the resulting balance between photosynthetic carbon (C) assimilation and the expenditure of fixed C by autotrophic respiration (Edwards *et al.* 1981; Waring & Running 1998). In forest ecosystems, respiration in above-ground woody tissues accounts for a significant portion of total autotrophic respiration (Edwards *et al.* 1981; Ryan *et al.* 1994; Ryan, Lavigne & Gower 1997); consequently, any reduction in woody-tissue respiration will increase the efficiency with which trees convert photosynthetically reduced C into new biomass. Bark photosynthesis provides a means for achieving such a reduction.

Stem photosynthetic activity has been reported in a wide variety of plant species and life forms (Schaedle 1975; Sprugel & Benecke 1991; Nilsen 1995). In woody trees, photosynthetic bark reduces the flux of CO₂ from woody tissues to the atmosphere; it does not result in net C gain on a whole-tissue basis. This process has been appropriately termed 'refixation' (Sprugel & Benecke 1991). Alternatively, the term 'corticular photosynthesis' may be applied because

the photosynthetic tissues are located in the bark cortex (Sprugel & Benecke 1991; Nilsen 1995). In contrast to woody trees, stems of many shrub and herb species have functional stomata which enable net C uptake from the atmosphere (e.g. Osmond *et al.* 1987; Comstock, Cooper & Ehleringer 1988; Comstock & Ehleringer 1990; Nilsen, Meinzer & Rundel 1989; Bossard & Rejmanek 1992; Nilsen *et al.* 1993). To distinguish it from refixation, this process is referred to as 'stem photosynthesis' (Nilsen 1995).

Investment of resources into refixation may provide trees with a means for improving their C balance that rivals investment in leaf photosynthesis. Equivalent gains in whole-plant C accumulation can be made either by increasing the rate of C uptake from the atmosphere (i.e. through leaf photosynthesis) or by decreasing respiratory C losses (i.e. through refixation). Refixation has an advantage over leaf photosynthesis in that there is very little associated water loss. This may be particularly important in temperate forests where seasonal drought regularly limits leaf photosynthesis. Moreover, because of the high CO₂ concentrations in woody tissues (Eklund 1990; Hari, Nygren & Korpilahti 1991; Eklund 1993; Levy

et al. 1999), refixation may have the added advantage of proceeding with a minimum of photorespiration. If plants invest nutrient resources where the C return per unit investment is maximized (Bloom, Chapin & Mooney 1985; Chapin *et al.* 1987), one would expect refixation to be favoured wherever bark surfaces transmit sufficient light for photosynthesis to occur.

Refixation may influence stand-level productivity by increasing C use efficiency, the ratio of net primary production to net photosynthesis. In boreal forests, Ryan *et al.* (1997) found that C use efficiency was greater in stands dominated by *Populus tremuloides* than in stands dominated by *Picea mariana* or *Pinus banksiana*. They suggested that greater C use efficiency in *Populus tremuloides* may be due to recycling of respired CO₂ through refixation. Instantaneous refixation rates in *Populus tremuloides* bark are known to reduce woody-tissue respiration by up to 90% (Foote & Schaedle 1976a; Foote & Schaedle 1978; Kharouk *et al.* 1995).

In this study, we analysed the functional controls on refixation in branches of Western White Pine (*Pinus monticola* Dougl. ex D. Don). We investigated effects of variation in temperature and light, in addition to morphological and physiological traits. We then combined the dominant physiological and environmental controls in a predictive model of refixation.

Materials and methods

SITE DESCRIPTION

Our study site was located in the University of Idaho's Experimental Forest (46°50' N, 116°43' W) at 915 m elevation. The climate at the site is typical of the inland Pacific North-west; it is characterized by cool, mild winters and warm, dry summers. Mean annual temperature is approximately 7 °C; mean annual precipitation is 720 mm. Soils at the site are deep, well-drained silt loams formed in volcanic ash deposited over granite residua. Two age classes of forest are present: a young plantation and a mature, naturally regenerated stand. This study made use of Western White Pine branches from trees growing in both the plantation and the mature stand.

GAS EXCHANGE MEASUREMENTS

Refixation is difficult to measure in woody tissues. Because of the small surface to volume ratios in these tissues, gas exchange remains in a non-steady state for 20–30 min following temperature or light perturbations (Foote & Schaedle 1976b; Brayman & Schaedle 1982). If exposed to solar radiation for this amount of time, transparent cuvettes accumulate heat (Sprugel & Benecke 1991). This is problematic because respiration and refixation are both sensitive to temperature and often respond differentially (Schaedle & Foote 1971; Foote & Schaedle 1976b; Han & Suzaki 1981; Brayman & Schaedle 1982). To avoid problems asso-

ciated with measuring respiration and refixation under fluctuating temperature conditions, we used a water bath to control cuvette air temperature (Mooney & Strain 1964; Foote & Schaedle 1976a,b; Han & Suzaki 1981; Brayman & Schaedle 1982).

A cuvette was constructed following the design of Foote & Schaedle (1976c). Water from a controlled-temperature bath circulated through a Plexiglas compartment attached to the front of a 1-l leaf cuvette (Li-Cor, Inc., Lincoln, NE); additionally, air inside the system passed through a coil of copper tubing submerged in the bath. This allowed us to control cuvette temperature over a range from 15 to 45 °C. A portable Li-6200 photosynthesis system (Li-Cor) measured CO₂ efflux. The Li-6200 was operated as an open system while gas exchange was in transient state to prevent CO₂ buildup and O₂ depletion within the cuvette; during steady-state observations the instrument was operated as a closed system.

The advantage of temperature control is often gained at the expense of working with excised tissue sections rather than whole tissues (e.g. Kriedemann & Buttrose 1971; Perry 1971; Schaedle & Foote 1971; Parker 1978; Coe & McLaughlin 1980; Linder & Troeng 1980; Levy & Jarvis 1998). Excision has the potential to alter CO₂ efflux in three ways: (1) by inducing wound respiration at the point of excision (Sprugel & Benecke 1991, and references therein); (2) by creating preferred pathways for CO₂ diffusion, and (3) by disrupting the transpiration stream, which may transport respired CO₂ either into or away from woody tissues (Negisi 1979; Hari *et al.* 1991; Martin, Teskey & Dougherty 1994; Sprugel *et al.* 1995; Levy *et al.* 1999).

To determine whether excision would influence respiration rates in our study, we compared *in situ* measurements of dark CO₂ efflux with excised tissue measurements on the same branches. Removable Plexiglas cuvettes were used to measure respiration on five branches on young trees in the plantation between 0900 and 1300 h on 9 July 1998. Cuvettes were attached to the sample branches using closed-cell foam gasket material and C-clamps (Cernusak *et al.* 2000). Sample branches were shaded for 30 min prior to measurement. Following measurement, the branches were excised and transported by automobile 56 km to the laboratory. Once in the laboratory, the cut surfaces were dipped in melted paraffin to minimize disruption of normal CO₂ diffusion pathways. Branch sections were placed in the cuvette individually; bark surface temperatures were matched to surface temperatures at the time of measurement in the field. This allowed us to compare *in situ* measurements to excised tissue measurements without Q_{10} adjustments to respiration rates. Diameters and lengths of measured branch sections ranged from 1.3 to 1.8 cm and from 7.5 to 11.7 cm, respectively. The controlled-temperature cuvette could accommodate branch sections up to 18 cm in length.

A paired *t*-test indicated that the excised-tissue respiration rates did not differ significantly from the *in situ* respiration rates ($P = 0.70$, $n = 5$). We concluded that our measurement protocol had not greatly altered branch gas exchange and all subsequent measurements were made on excised branch sections in the controlled-temperature cuvette.

LIGHT AND TEMPERATURE RESPONSES

A ceptometer (Decagon Devices, Pullman, WA) was used to determine the branch exposure that was likely to intercept the most photosynthetically active radiation (PAR). The ceptometer was held parallel to branches growing in each of the four cardinal directions on five trees in the plantation. These measurements indicated that branches on the north and east sides of trees were likely to intercept more PAR than branches on the south and west sides. Branches were therefore selected from north-east exposures for light response measurements. Ten branches were measured, each from a different young tree in the plantation. Western White Pine generally has 3 years of needle retention. We sampled 4-year-old branches (fourth whorl from tree top) to ensure at least one needle-free internode on which to measure gas exchange. Sample branches were collected in the early morning and gas exchange measurements were completed on the same day.

Respiration was first measured in the dark, and then incrementally at irradiances of 100, 250, 500, 1000 and 2000 $\mu\text{mol PAR m}^{-2} \text{s}^{-1}$. An artificial light source (Quantum Devices, Inc., Barneveld, WI) provided PAR at the desired intensities. Branches were illuminated from one direction and were oriented in the cuvette so that the top of the branch faced the light source. Bark surface temperatures on the illuminated sides of branches were maintained at $25 \pm 2^\circ\text{C}$. CO_2 efflux was measured six times on each branch at each irradiance. Light response measurements were completed between 15 and 22 July 1998.

Repeated measures analysis was used to test for significant differences in light response among branches (Potvin, Lechowicz & Tardif 1990). Replicated gas exchange measurements at each light intensity were used to compare branches. This analysis indicated that between-branch differences were significant. This being the case, the following equation was fitted to each individual branch to obtain an estimate of the light-saturated rate of gross photosynthesis and the initial slope of the light response curve (Landsberg 1977):

$$P_{g_i} = P_{g_{\max}} \{1 - \exp[-\alpha(I)]\}, \quad \text{eqn 1}$$

where P_{g_i} is instantaneous gross photosynthesis, calculated as the difference between CO_2 efflux in the dark and that in the light, $P_{g_{\max}}$ is the light-saturated rate of gross photosynthesis, α is the initial slope of the light response curve, or quantum efficiency, and I is the amount of PAR reaching the bark surface.

Non-linear regression was used to estimate the $P_{g_{\max}}$ and α parameters.

The temperature response of refixation was measured on six branches collected from young trees growing in the plantation. Branches were collected following the protocol established for light response measurements. Respiration was measured at bark surface temperatures of 15, 25, 35 and $45 \pm 2^\circ\text{C}$. Gas exchange was measured in the dark and at 1000 $\mu\text{mol PAR m}^{-2} \text{s}^{-1}$ at each temperature. Temperature response measurements were completed between 27 July and 4 August 1998.

SURFACE CONDUCTANCE AND INTERNAL CO_2 CONCENTRATION

To measure the response of refixation to variation in surface conductance, bark chlorophyll concentration, bark nitrogen (N) concentration and specific bark area, branches were collected from three crown positions in the mature stand. Sectional climbing ladders were used to access the lower, middle and upper crowns of three trees. The trees measured 21, 23 and 24 cm in diameter at breast height and were approximately 20 m tall. A total of 27 branches were collected, three from each crown position on each tree, all from north-east exposures. Branches increased in diameter and age from the upper crown to the lower crown and thus represented all branches on the tree. Refixation was estimated from CO_2 efflux in the dark (R_d) and under illumination with 1000 $\mu\text{mol PAR m}^{-2} \text{s}^{-1}$ (R_{lt}) as follows:

$$\% \text{ refixation} = \left(\frac{R_d - R_{\text{lt}}}{R_d} \right) \times 100. \quad \text{eqn 2}$$

Bark surface temperature was maintained at $25 \pm 2^\circ\text{C}$ during all measurements. All gas exchange observations were completed on the same day on which branches were collected. Measurements took place between 10 and 13 August 1998.

Immediately following gas exchange measurements, the branches were sectioned. One half was used for surface conductance measurements; the other was frozen and stored for measurements of chlorophyll, N and specific bark area. For surface conductance measurements, branch sections were weighed to the nearest 0.1 mg. They were then exposed to wind speeds of approximately 2 m s^{-1} while relative humidity and air temperature were monitored. After 24 h, water loss was determined by re-weighing each branch. The temperature and humidity measurements were averaged over the 24-h period. Total vapour conductance of the bark surface was calculated as (Campbell & Norman 1998)

$$g_v = \frac{E}{(C_{\text{vs}} - C_{\text{va}})}, \quad \text{eqn 3}$$

where g_v is the vapour conductance of the bark plus the boundary layer, E is the water loss over the 24-h measurement period, C_{vs} is the vapour concentration at the surface, and C_{va} is the vapour concentration

of the atmosphere. The vapour concentration of air beneath the bark surface was assumed to be saturated; that of the atmosphere was calculated from the relative humidity and temperature data. Boundary layer conductance for each branch section was determined as (Campbell & Norman 1998)

$$g_{va} = 0.147 \sqrt{u/d}, \quad \text{eqn 4}$$

where g_{va} is the boundary layer conductance, u is the wind speed, and d is the characteristic dimension. The diameter of each branch was used as its characteristic dimension (Campbell & Norman 1998). Boundary layer conductance was subtracted from total conductance to calculate bark surface vapour conductance. This was then converted to CO_2 conductance by multiplying by 0.66 (Campbell & Norman 1998). Surface conductances and CO_2 efflux rates were used to estimate CO_2 concentrations beneath bark surfaces during gas exchange measurements:

$$F = g_c(C_i - C_a), \quad \text{eqn 5}$$

where F is the CO_2 efflux from the branch, g_c is the conductance of the bark surface to CO_2 , C_a is the concentration of CO_2 in the atmosphere, and C_i is the concentration of CO_2 beneath the bark surface. Boundary layer resistance in the cuvette was assumed to be negligible; C_a was assumed to be $365 \mu\text{mol mol}^{-1}$.

Cortical CO_2 concentrations were measured on young trees in the field to confirm that our calculated values were representative of actual values. Our method for collecting CO_2 samples was similar to that used by Eklund (1990, 1993) and Eklund *et al.* (1992). Glass tubes coupled to hypodermic needles were inserted into the bark of 12 branches in the plantation. Each hypodermic needle penetrated a septum on a 2-ml sample vial. The gas concentration in the vial was allowed to equilibrate with the air beneath the bark surface for several hours; the vial was then removed and its contents measured for CO_2 concentration by gas chromatography (Model 5730 A, Hewlett Packard, Palo Alto, CA). This method was used to measure cortical CO_2 concentrations in the absence of direct solar radiation at dawn, and again in the afternoon on 19 August 1998.

SPECIFIC BARK AREA, BARK CHLOROPHYLL CONCENTRATION AND BARK N CONCENTRATION

A large increment borer was used to remove 1-cm² pieces of bark from branches of both young and mature trees on which gas exchange had been measured. Chlorophyll concentrations were determined by macerating the bark samples in buffered 80% aqueous acetone; extractions were performed on ice and in semi-darkness. Extract absorbances were measured with a spectrophotometer (Model DU 640,

Beckman Instruments, Inc., Fullerton, CA); total chlorophyll contents and chlorophyll *a/b* ratios were calculated according to Porra *et al.* (1989).

A second 1-cm² bark section was removed from each branch for determination of specific bark area and bark N concentration. The bark chips were dried for 48 h at 70 °C and weighed to the nearest 0.1 mg. They were then ground to a fine powder and analysed for N contents using an elemental analyser (CHN 600, Leco Corporation, St. Joseph, MI).

ANALYSIS

Linear correlation was used to detect relationships between measured morphological and physiological parameters and gross photosynthesis in bark. For some variables, values expressed per unit mass yielded stronger correlations than values expressed per unit area. In these cases, we used the specific bark area for each branch to make the conversion from area-based units to mass-based units. Linear regression was used to develop predictive relationships between measured parameters and gross photosynthesis. In the case of a non-significant intercept, the regression line was forced through the origin and R^2 was calculated according to equation 1 of Kvålseth (1985). Analysis of variance was used to detect differences in gross photosynthesis and related traits among crown positions in the mature trees. Where necessary, data were transformed to meet the assumptions of normality and equal variance. If significant differences were detected among crown positions, Tukey's method for pair-wise comparisons was used to determine which crown positions differed. All statistical analyses were performed in SYSTAT 7.0 (SPSS Inc., Chicago, IL).

Results

LIGHT AND TEMPERATURE RESPONSES

Gross photosynthesis increased with light intensity to $> 2000 \mu\text{mol PAR m}^{-2} \text{ s}^{-1}$. The response of gross photosynthesis to light intensity differed among branches ($P < 0.0001$, $n = 10$). Fitting of equation 1 to data from all branches resulted in an average light response with a quantum efficiency of $0.0014 \mu\text{mol CO}_2 \mu\text{mol}^{-1}$ photons, and a light-saturated gross photosynthetic rate of $1.61 \mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$ (Fig. 1a). Refixation expressed as a percentage of dark respiration ranged from 65 to 90% at $2000 \mu\text{mol PAR m}^{-2} \text{ s}^{-1}$, with a mean value of 76% (Fig. 1b).

The average proportional increase in dark respiration with each 10 °C increase in temperature (Q_{10}) was 1.70 ± 0.07 (mean $\pm$ SE). Gross photosynthesis increased over the temperature range from 15 to 45 °C; however, it did not increase as rapidly as dark respiration (Fig. 2a). Thus, the percentage of dark-respired CO_2 that was refixed declined with increasing bark surface temperature (Fig. 2b).

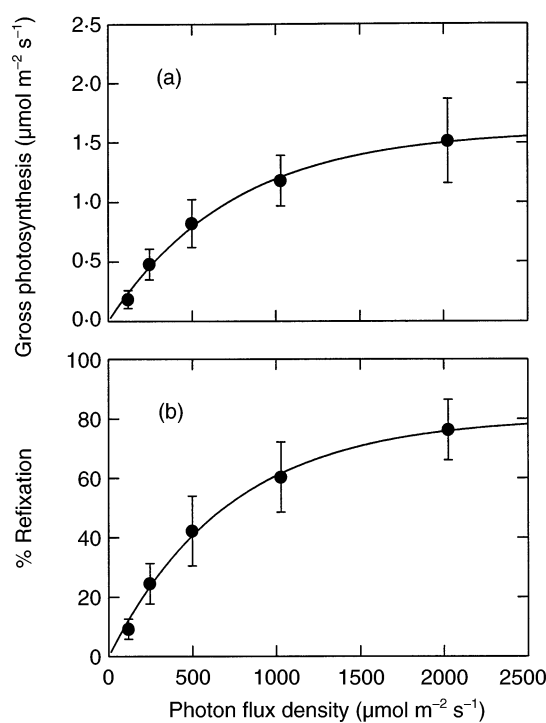


Fig. 1. The average light response of branches of young Western White Pine expressed as (a) gross photosynthesis, calculated as the difference between dark and light respiration rates, and (b) refixation as a percentage of dark respiration. Measurements were made at bark surface temperatures of 25 ± 2 °C. Error bars represent standard deviations; $n = 10$.

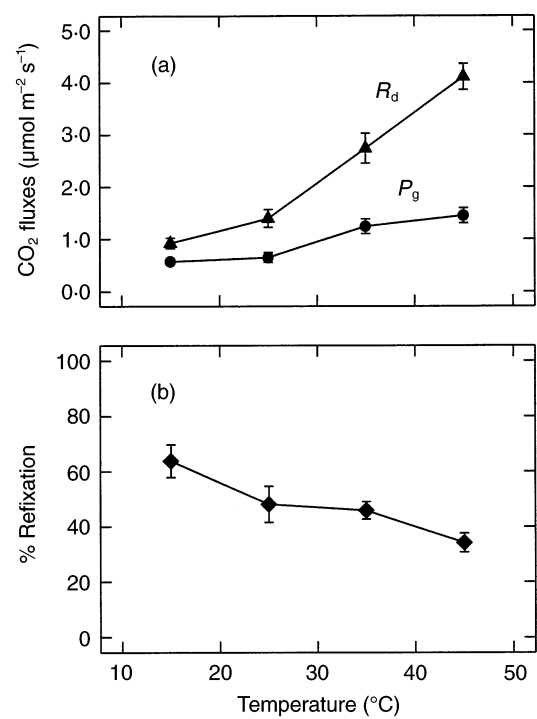


Fig. 2. Temperature response of branches of young trees for (a) dark respiration (R_d) and gross photosynthesis (P_g), gross photosynthesis being measured at $1000 \mu\text{mol PAR m}^{-2} \text{ s}^{-1}$, and (b) refixation as a percentage of dark respiration. Error bars represent standard errors; $n = 6$.

SURFACE CONDUCTANCE AND INTERNAL
CO₂ CONCENTRATION

The mean bark surface conductance to water vapour was $1.03 \text{ mmol m}^{-2} \text{ s}^{-1}$. The corresponding mean CO₂ conductance was $0.68 \text{ mmol m}^{-2} \text{ s}^{-1}$. The mean boundary layer vapour conductance was $1.6 \text{ mol m}^{-2} \text{ s}^{-1}$;

more than 3 orders of magnitude larger than the mean bark surface vapour conductance. Bark surface conductance was negatively correlated to gross photosynthesis (Table 1).

A strong, positive relationship existed between gross bark photosynthesis and internal CO₂ concentration in the dark (Fig. 3a, Table 1). In contrast, only a weak

Table 1. Pearson correlation coefficients (R) and associated probabilities (P) for gross photosynthesis in Western White Pine branches and various morphological and physiological parameters. Gross photosynthesis was measured at $1000 \mu\text{mol PAR m}^{-2} \text{ s}^{-1}$ and 25 ± 2 °C. Physiological parameters that are expressed in mass-based units are correlated with mass-based gross photosynthesis. Area-based gross photosynthesis was converted to mass-based gross photosynthesis using specific bark area. Except where noted, measurements were made on branches of both young and mature trees

Parameter	Gross photosynthesis/area ($\mu\text{mol CO}_2 \text{ m}^{-2} \text{ bark s}^{-1}$)		Gross photosynthesis/mass ($\mu\text{mol CO}_2 \text{ kg}^{-1} \text{ bark s}^{-1}$)	
	R	P	R	P
Dark respiration ($\mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$)	0.89	< 0.0001		
Dark internal CO ₂ concentration* ($\mu\text{mol mol}^{-1}$)	0.87	< 0.0001		
Light internal CO ₂ concentration* ($\mu\text{mol mol}^{-1}$)	0.35	NS†		
Bark surface conductance* ($\text{mmol H}_2\text{O m}^{-2} \text{ s}^{-1}$)	-0.62	< 0.001		
Specific bark area ($\text{m}^2 \text{ kg}^{-1}$)			0.76	< 0.0001
Bark N concentration (mmol m^{-2})	-0.23	NS		
Bark N concentration (mg g^{-1})			0.38	< 0.05
Bark chlorophyll concentration ($\mu\text{mol m}^{-2}$)	-0.13	NS		
Bark chlorophyll concentration ($\mu\text{mol g}^{-1}$)			0.34	< 0.05
Bark Chl/N ratio ($\mu\text{mol mmol}^{-1}$)	0.03	NS		
Chlorophyll <i>a/b</i> ratio ($\mu\text{mol } \mu\text{mol}^{-1}$)	0.44	< 0.01		

*Measured on branches of mature trees only.
†NS = not significant at $P \leq 0.05$.

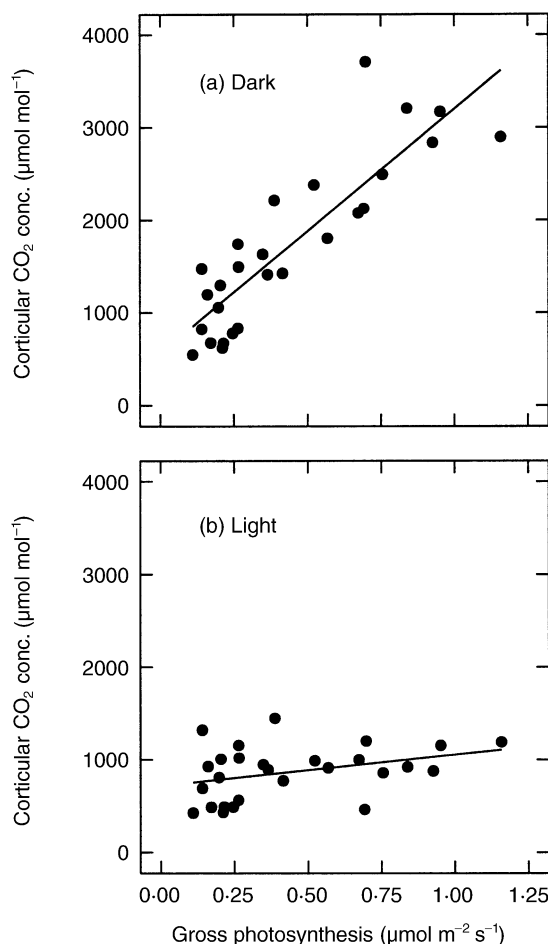


Fig. 3. Gross photosynthesis measured in the light vs calculated internal CO_2 concentration (a) in the dark, and (b) in the light. Internal CO_2 concentrations were solved for using bark surface conductance and flux rates measured on branches of mature trees. Measurements were made in the dark and at $1000 \mu\text{mol PAR m}^{-2} \text{s}^{-1}$ with bark surface temperatures of $25 \pm 2^\circ\text{C}$. Correlation statistics are (a) $R = 0.87$, $P < 0.0001$, $n = 27$, and (b) $R = 0.35$, $P = 0.08$, $n = 27$.

relationship existed between internal CO_2 concentration in the light and gross photosynthesis (Fig. 3b, Table 1). Internal CO_2 concentrations were substantially higher in the dark than in the light (Table 2).

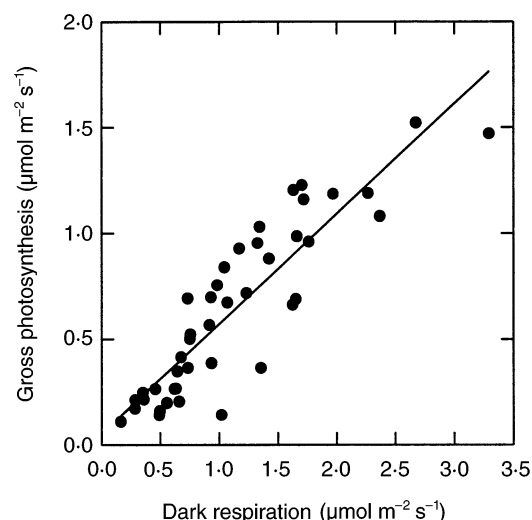


Fig. 4. The relationship between gross photosynthesis and dark respiration in branches of young and mature trees combined. Both parameters were measured at bark surface temperatures of $25 \pm 2^\circ\text{C}$. Gross photosynthesis was measured at $1000 \mu\text{mol PAR m}^{-2} \text{s}^{-1}$.

The mean cortical CO_2 concentration measured *in situ* was $992 \pm 36 \mu\text{mol mol}^{-1}$ (mean $\pm$ SE) at 0800 h on 19 August 1998. At 1500 h on the same day, the mean CO_2 concentration in the bark had increased to $1693 \pm 190 \mu\text{mol mol}^{-1}$ (mean $\pm$ SE). The dawn sample is based on 12 measured branches and the afternoon sample on 11 measured branches. The minimum and maximum air temperatures on the day that measurements were taken were 2 and 28°C , respectively, at a nearby weather station in Potlatch, ID.

RELATIONSHIPS BETWEEN REFIXATION AND MORPHOLOGICAL AND PHYSIOLOGICAL PARAMETERS

We found a striking correspondence between whole-tissue dark respiration and bark photosynthesis (Table 1). A strong linear relationship existed over the range of dark respiration rates (Fig. 4). Variation in whole-tissue dark respiration explained 79% of the variation

Table 2. Descriptive statistics for morphological and physiological traits of Western White Pine branches. Dark respiration and gross photosynthesis were measured at $25 \pm 2^\circ\text{C}$. Photosynthesis was measured at $1000 \mu\text{mol PAR m}^{-2} \text{s}^{-1}$

Parameter	<i>n</i>	Mean	Standard deviation
Branch diameter (cm)	43	1.68	0.37
Dark respiration ($\mu\text{mol CO}_2 \text{ m}^{-2} \text{s}^{-1}$)	43	1.14	0.69
Gross photosynthesis ($\mu\text{mol CO}_2 \text{ m}^{-2} \text{s}^{-1}$)	43	0.64	0.40
Bark surface conductance* ($\text{mmol H}_2\text{O m}^{-2} \text{s}^{-1}$)	27	1.03	0.38
Calculated dark internal CO_2 concentration* ($\mu\text{mol mol}^{-1}$)	27	1726	897
Calculated light internal CO_2 concentration* ($\mu\text{mol mol}^{-1}$)	27	869	286
Specific bark area ($\text{m}^2 \text{kg}^{-1}$)	43	1.45	0.24
Bark N concentration (mmol m^{-2})	43	245	55
Bark chlorophyll concentration ($\mu\text{mol m}^{-2}$)	43	404	102
Bark Chl/N ratio ($\mu\text{mol mmol}^{-1}$)	43	1.71	0.55
Bark chlorophyll <i>a/b</i> ratio ($\mu\text{mol } \mu\text{mol}^{-1}$)	43	1.67	0.59

*Measured on branches of mature trees only.

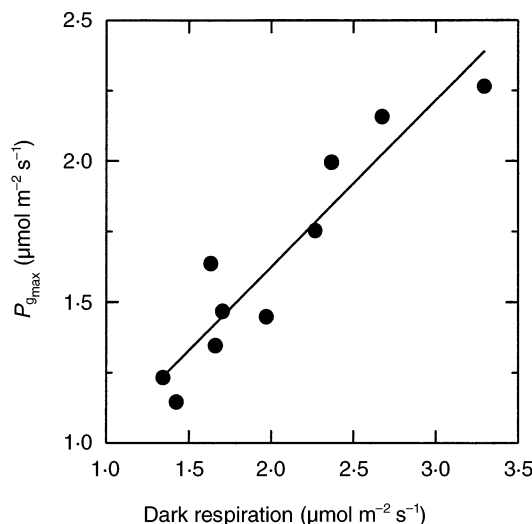


Fig. 5. Light-saturated gross photosynthesis ($P_{g\max}$) as a function of dark respiration in 4-year-old branches from young trees. Maximum rates were solved for asymptotically using non-linear regression. All measurements were made at bark surface temperatures of $25 \pm 2^\circ\text{C}$.

in bark photosynthesis ($R^2 = 0.79$, $P < 0.0001$, $n = 43$). The slope of the relationship between the two parameters was 0.55 and the intercept did not differ significantly from zero ($P = 0.34$). This suggests that, across the range of dark respiration rates observed in this study, photosynthetic bark reduced whole-tissue dark respiration rates by 55% when branches were illuminated with $1000 \mu\text{mol PAR m}^{-2} \text{s}^{-1}$, and with bark surface temperatures maintained at $25 \pm 2^\circ\text{C}$.

Dark respiration remained closely related to gross photosynthesis at light-saturated rates in the 10 branches measured for light response ($R = 0.94$, $P < 0.0001$, $n = 10$; Fig. 5). When the intercept was forced through the origin, dark respiration rates explained 77% of the variation in gross photosynthesis and the slope of the relationship was 0.79 ($R^2 = 0.77$, $P < 0.0001$, $n = 10$); thus, a refixation rate at light saturation of 79% is predicted.

Gross photosynthesis was strongly correlated with specific bark area (Table 1, Fig. 6). Neither bark N concentration nor bark chlorophyll concentration correlated with gross photosynthesis when expressed per unit area. However, conversion of these variables to mass-based units resulted in a significant correlation in both cases (Table 1). For these analyses, we also converted gross photosynthesis to a mass basis, analogous to the method of Field & Mooney (1986).

Among the other measured traits, bark chlorophyll *a/b* ratio correlated significantly with gross photosynthesis per unit area (Table 1). The chlorophyll/N ratio in the bark was not correlated with gross photosynthesis. Descriptive statistics for observations of specific bark area, bark chlorophyll concentration, bark N concentration, chlorophyll *a/b* ratio and chlorophyll/N ratio are presented in Table 2.

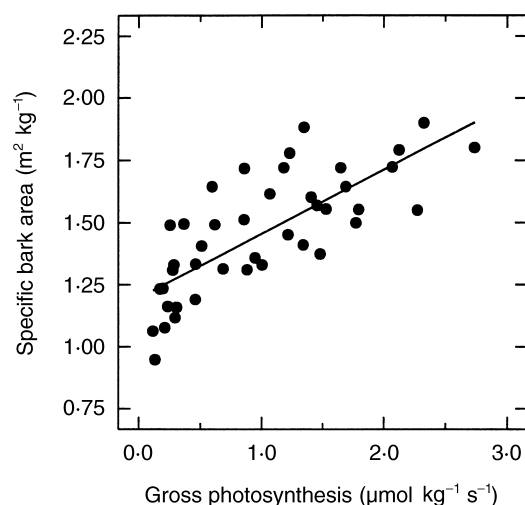


Fig. 6. The relationship between gross photosynthesis and specific bark area in branches of young and mature trees combined. Photosynthesis is expressed as $\mu\text{mol CO}_2 \text{ kg}^{-1} \text{ bark s}^{-1}$ to make it compatible with specific bark area. Photosynthesis was measured at $1000 \mu\text{mol PAR m}^{-2} \text{s}^{-1}$ and bark surface temperatures of $25 \pm 2^\circ\text{C}$. Correlation statistics are $R = 0.76$, $P < 0.0001$, $n = 43$.

VARIATION IN MEASURED PARAMETERS AMONG CROWN POSITIONS

We observed distinct changes in gas exchange parameters with increasing depth in the crown of mature Western White Pine (Table 3, Fig. 7a,b). Both respiration and refixation decreased continuously with canopy depth as branches became larger and older. Ninety per cent of the variation in refixation rates was accounted for by considering crown position and individual variance (Table 3). Of the two, crown position variance accounted for most of the variation in gross photosynthetic rates; this was also true of respiration (Table 3). Branch respiration rates were more than 2-fold faster in the upper crown than in the lower crown when expressed on a surface area basis, and more than 3-fold faster when expressed on a total volume basis.

Static branch traits varied with crown position (Table 3, Fig. 7). Specific bark area decreased markedly with increasing canopy depth, as did dark CO_2 concentrations. Variation in dark CO_2 concentrations partly resulted from variation in bark conductance, which increased with increasing canopy depth. Bark N concentration decreased with increasing canopy depth when expressed per unit bark mass, but did not vary when expressed per unit bark area ($P = 0.57$). Bark chlorophyll *a/b* ratio decreased with increasing canopy depth. Neither bark chlorophyll concentration nor chlorophyll/N ratio varied among crown positions.

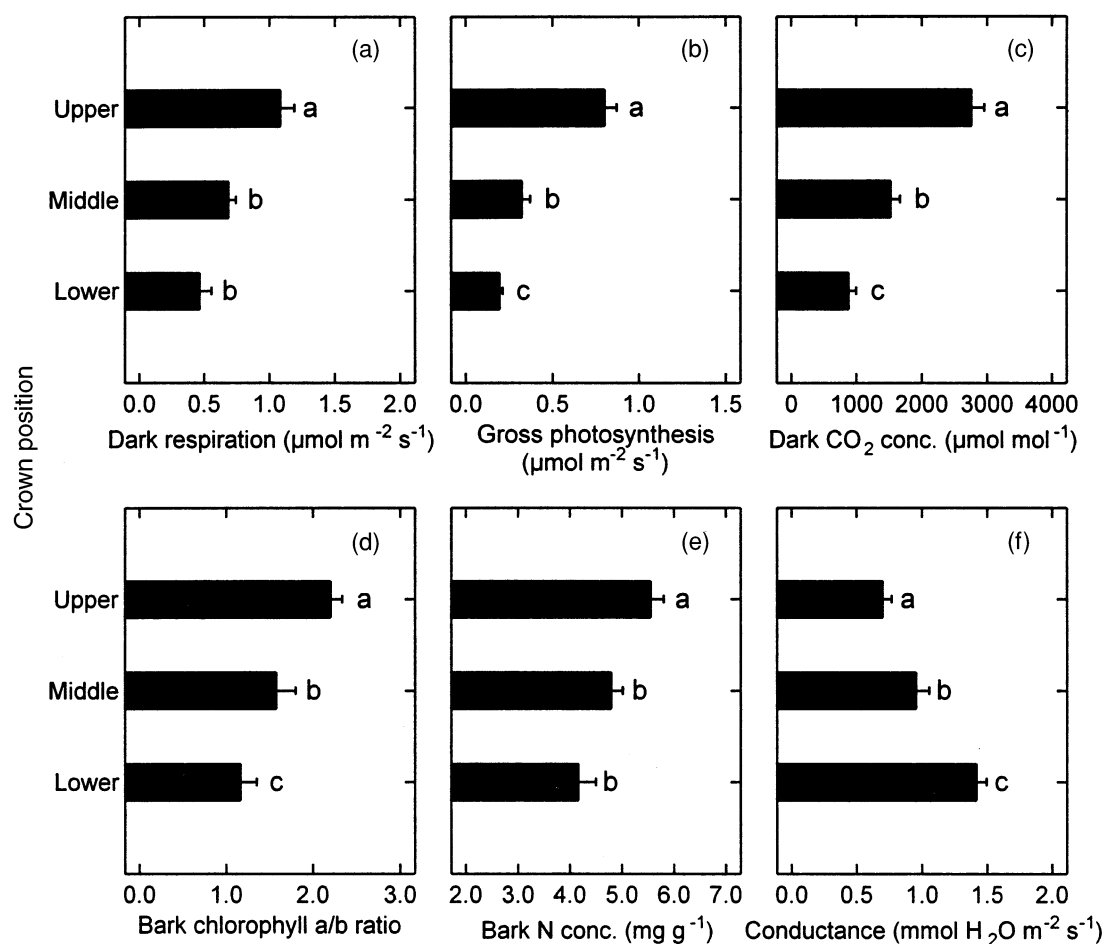
Discussion

The capacity of Western White Pine bark for photosynthetic refixation is striking. We observed a 55%

Table 3. Analysis of variance (ANOVA) results for morphological and physiological traits of branches from mature Western White Pine trees as related to crown position. Three crown positions from each of three trees were sampled; $n = 27$ for each analysis

	F-ratio			
	Crown position	Individual	Crown position $\times$ individual	Model R^2
Dark respiration ($\mu\text{mol m}^{-2} \text{s}^{-1}$)	19.18 ****	3.74 *	1.54 NS	0.74
Gross photosynthesis ($\mu\text{mol m}^{-2} \text{s}^{-1}$)	67.64 ****	6.00 *	2.06 NS	0.90
Specific bark area ($\text{m}^2 \text{kg}^{-1}$)	6.59 **	3.54 NS	2.68 NS	0.63
Bark chlorophyll concentration ($\mu\text{mol m}^{-2}$)	0.48 NS	7.66 **	1.92 NS	0.57
Chlorophyll <i>a/b</i> ratio ($\mu\text{mol } \mu\text{mol}^{-1}$)	25.18 ****	18.53 ****	2.60 NS	0.84
Bark N concentration (mg g^{-1})	12.16 ***	6.51 **	2.10 NS	0.72
Chl/N ratio ($\mu\text{mol mmol}^{-1}$)	0.79 NS	15.39 ***	1.81 NS	0.69
Bark conductance ($\text{mmol H}_2\text{O m}^{-2} \text{s}^{-1}$)	29.93 ****	5.17 *	1.02 NS	0.80
Dark internal CO_2 concentration ($\mu\text{mol mol}^{-1}$)	49.57 ****	2.88 NS	0.76 NS	0.86
Light internal CO_2 concentration ($\mu\text{mol mol}^{-1}$)	6.27 **	6.43 **	1.61 NS	0.64

NS, $P > 0.05$; * $P \leq 0.05$; ** $P < 0.01$; *** $P < 0.001$; **** $P < 0.0001$.

**Fig. 7.** The distribution of selected morphological and physiological branch traits within the crown of mature Western White Pine. Bars represent means; error bars represent 1 standard error. Bars within a frame followed by different letters are significantly different at $P \leq 0.05$. For each crown position, $n = 9$.

reduction in CO_2 efflux upon illumination with $1000 \mu\text{mol PAR m}^{-2} \text{s}^{-1}$; the reduction was seen in every measured branch, including branches collected at depth in the canopy. Temperature and irradiance controlled refixation directly. At a given temperature and irradiance, the strongest correlation was with dark

respiration; the strongest correlation with a morphological parameter was with specific bark area (bark area per unit mass).

Refixation in Western White Pine was not saturated at light intensities up to $2000 \mu\text{mol PAR m}^{-2} \text{s}^{-1}$. This light response contrasts with observations in

Fagus crenata, *Quercus acutissima* (Han & Suzaki 1981), *Populus tremuloides* (Foote & Schaedle 1976b) and *Acer rubrum* (Coe & McLaughlin 1980); in these species bark photosynthesis saturated between 200 and 300 $\mu\text{mol PAR m}^{-2} \text{s}^{-1}$. However, the light response of Western White Pine is similar to that of a sahelian shrub species, *Guiera senegalensis*, in which refixation increased linearly to near 2000 $\mu\text{mol PAR m}^{-2} \text{s}^{-1}$ and apparently never saturated (Levy & Jarvis 1998). Species-level differences in light response may be due to differences in periderm transmissivity. On average, about 15% of incident PAR penetrates the periderm and becomes available for corticular photosynthesis; however, this may vary greatly with age, species and environment (Schaedle 1975).

The suggestion of low bark transmissivity in Western White Pine is further supported by the chlorophyll *alb* ratios (Table 2); these ratios are even lower than those typically found in shade-acclimated leaves (Lichtenthaler *et al.* 1981; Murchie & Horton 1997). However, chlorophyll *alb* ratios in the bark of Western White Pine are similar to those measured in the bark of other tree species (Pearson & Lawrence 1958; Larcher *et al.* 1988; Kauppi 1991; Kharouk *et al.* 1995; Solhaug, Gauslaa & Haugen 1995).

Specific bark area is an integrated measure of bark thickness and density. As specific bark area decreases, penetration of sunlight into the cortex may also decrease. This could explain the observed correlation between specific bark area and gross photosynthetic rates. It is interesting to note the conceptual similarity between this observation and recent reports of variation in leaf photosynthetic rates in relation to specific leaf area (e.g. Reich, Ellsworth & Walters 1998).

Maximum gross photosynthesis in the bark of Western White Pine ranged from 1.1 to 2.3 $\mu\text{mol CO}_2 \text{m}^{-2} \text{s}^{-1}$ when expressed on a total surface area basis. If expressed on a projected area basis, maximum gross photosynthesis ranged from 3.6 to 7.1 $\mu\text{mol m}^{-2} \text{s}^{-1}$. These rates are similar to maxima observed for *Populus tremuloides* branches under isotropic illumination (5.5–7.5 $\mu\text{mol m}^{-2} \text{s}^{-1}$) (Foote & Schaedle 1978; Brayman & Schaedle 1982). Expressed on a projected area basis, our maximum refixation rates fall within the range of photosynthetic rates reported for foliage of *Pinus strobus* (Eastern White Pine) (Maier & Teskey 1992; Gower, Reich & Son 1993).

Gross photosynthesis increased over a range of bark surface temperatures from 15 to 45 °C (Fig. 2a). This suggests an exceptionally high temperature optimum for photosynthesis in the bark of Western White Pine (Larcher 1995). We believe this observation can best be explained by the high CO₂ concentrations within the bark. Temperature optima for C₃ photosynthesis are commonly between 20 and 30 °C (Larcher 1995; Nilsen 1995). However, at high CO₂ concentrations, temperature optima are expected to increase because CO₂ inhibits photorespiration

(Larcher 1995). In the laboratory, we estimated CO₂ concentrations of $869 \pm 55 \mu\text{mol mol}^{-1}$ (mean $\pm$ SE) within illuminated branches. Concentrations measured in the field were $992 \pm 36 \mu\text{mol mol}^{-1}$ at 0800 h and $1693 \pm 190 \mu\text{mol mol}^{-1}$ at 1500 h. Although many authors have speculated that high CO₂ concentrations are an important feature of refixation (e.g. Schaedle 1975; Foote & Schaedle 1976b; Levy & Jarvis 1998), our results are the first to confirm this suggestion. We speculate that high photosynthetic temperature optima are a general characteristic of refixation among taxa due to the high CO₂ concentrations in refixing tissues.

Dark respiration emerged from our analysis as the strongest correlate with gross photosynthesis when temperature and irradiance were held constant. In branches collected near the bottom of the canopy, dark respiration and gross photosynthesis were depressed in concert (Fig. 7a,b). Consequently, the percentage reduction in CO₂ efflux upon illumination was fairly constant over a broad range of tissue ages and canopy microenvironments (Fig. 4). Proportional decreases in respiration and refixation with stem section age have also been observed in *Pinus sylvestris* (Linder & Troeng 1980), *Fagus crenata*, *Quercus acutissima* (Han & Suzaki 1981) and *Populus tremuloides* (Schaedle & Foote 1971; Brayman & Schaedle 1982). Moreover, measurements on a single *Populus tremuloides* tree over the course of an entire year revealed striking correspondence between seasonal changes in dark respiration and seasonal changes in refixation when the two were measured at constant temperature (Foote & Schaedle 1976a). In the present study, we observed the same correlation between refixation and dark respiration rates even within a single age class (Fig. 5). The correlation between refixation rates and dark respiration rates appears to be more general than the correlation with tissue age.

Because no mechanism is readily apparent, it seems unlikely that dark respiration causes variation in refixation rates. We did observe a strong correlation between internal CO₂ concentration in the dark and refixation rates (Fig. 3a). However, if CO₂ is a direct control on refixation, the concentration during illumination should be more important than the dark concentration. We failed to observe a significant correlation between refixation and CO₂ concentrations in the light (Fig. 3b). These results lead us to suggest, instead, that trees allocate their photosynthetic capacity for refixation relative to dark respiration rates. This would result in efficient use of N for recycling respired CO₂, and would be predicted from current paradigms of resource allocation patterns within plants (Bloom *et al.* 1985; Chapin *et al.* 1987).

Efficient deployment of N resources within a canopy should yield similar rates of C fixation per unit of N invested in photosynthetic tissues. Photosynthetic N use efficiency (PNUE), the maximum

photosynthetic rate per unit N, provides a means of testing this hypothesis. Using gross photosynthetic rates expressed on a projected area basis, we estimated maximum PNUE of $24.0 \pm 1.6 \mu\text{mol CO}_2 \text{ mol}^{-1} \text{ N s}^{-1}$ in the bark of Western White Pine branches. This compares favourably with estimates of PNUE based on net photosynthetic rates for the foliage of *Pinus strobus* (24.1) and *Pinus resinosa* (23.7) (Gower *et al.* 1993).

Whereas PNUE of bark is comparable to that of foliage, water-use efficiency (WUE) is much greater. Instantaneous WUE is typically defined as the ratio of net photosynthesis to transpiration; for bark, we defined it as the ratio of gross photosynthesis to transpiration. We measured a vapour pressure deficit (VPD) of approximately 0.018 in the laboratory. Using this VPD and maximum photosynthetic rates, the WUE of Western White Pine bark was estimated as $131 \pm 10 \text{ mmol CO}_2 \text{ mol}^{-1} \text{ H}_2\text{O}$. This value can be compared to published values of WUE for Western White Pine foliage. Marshall & Zhang (1994) reported a value of 11.8 Pa for $P_a - P_i$, the difference in CO_2 partial pressure between the atmosphere and leaf intercellular spaces, in foliage of Western White Pine. Using their equation 2 and our laboratory VPD, we estimated a foliar WUE of $2.7 \text{ mmol CO}_2 \text{ mol}^{-1} \text{ H}_2\text{O}$. This is approximately 50-fold smaller than the estimate of WUE for bark. The discrepancy underscores a fundamental difference between photosynthesis in leaves and bark: leaves must expose moist tissues to the atmosphere to take up CO_2 , whereas bark need not expose moist tissues to take up CO_2 respired by underlying heterotrophic tissues.

Although resource allocation theory would suggest that the distribution of refixation capacity can be predicted on the basis of N and water constraints, we found a simpler and apparently more general (albeit more empirical) predictive relationship: the relationship between dark respiration and refixation. We used this relationship to model refixation. We present the following model:

$$P_g = \{1 - \exp[-\alpha(I)]\} \times \left(\frac{P_{g_{\max}}}{P_{g_{1000}}} \right) \times (Q_{10P_g}^{(T-25)/10}) \times 0.55 \times \left(\frac{R_T}{Q_{10R_d}^{(T-25)/10}} \right), \quad \text{eqn 6}$$

where P_g is gross photosynthesis in the bark, α is quantum efficiency, I is incident PAR intensity, $P_{g_{\max}}$ is light-saturated gross photosynthesis, $P_{g_{1000}}$ is the gross photosynthetic rate at $1000 \mu\text{mol PAR m}^{-2} \text{ s}^{-1}$, Q_{10P_g} is the average increase in gross photosynthesis with a 10°C increase in temperature, T is the bark surface temperature, R_T is dark respiration at temperature T , and Q_{10R_d} is the average increase in dark respiration with a 10°C increase in temperature. The first term in the equation is a light modifier based on the light response curve; $P_{g_{\max}}/P_{g_{1000}}$ adjusts the light modifier so that it equals unity at $1000 \mu\text{mol PAR m}^{-2} \text{ s}^{-1}$; the third term is a temperature modifier for

gross photosynthesis in exponential form; 0.55 is the slope of the line yielded by regression of gross photosynthesis on dark respiration at $1000 \mu\text{mol PAR m}^{-2} \text{ s}^{-1}$ and 25°C , and the last term adjusts the dark respiration rate for temperatures other than 25°C . Parametrization of the model using the results obtained for Western White Pine yields

$$P_g = \{1 - \exp[-0.0014(I)]\} \times 1.35 \times (1.58^{(T-25)/10}) \times 0.55 \times \left(\frac{R_T}{1.80^{(T-25)/10}} \right), \quad \text{eqn 7}$$

which predicts gross photosynthesis using temperature, light intensity and dark respiration as input variables. The Q_{10} values in this equation are based on temperature responses between 15 and 35°C . Over this temperature range, both gross photosynthesis and dark respiration can be described by exponential relationships. At 45°C , gross photosynthesis departs from the exponential curve (Fig. 2a), possibly due to thermal degradation of enzymes. Regardless of the source of departure, it is unlikely that bark surface temperatures will approach 45°C in the naturally occurring range of Western White Pine.

Only a few authors have attempted to incorporate refixation into whole-plant or ecosystem C budgets (Foote & Schaedle 1978; Benecke 1985; Levy & Jarvis 1998). In part, this may reflect a historical deficiency in research directed at understanding the gas exchange of woody tissues. However, current interest in C budgets at all scales has resulted in improved methods for measuring and predicting dark respiration in woody tissues. On the basis of results obtained in this study, and a reconsideration of previously published data, it appears that there is a strong, empirical relationship between dark respiration in woody tissues and the capacity for refixation in the surrounding bark. This relationship will require further investigation before it can be wholly understood. In the meantime, we have provided a means of estimating refixation at whole-plant and ecosystem scales from existing models of dark respiration.

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